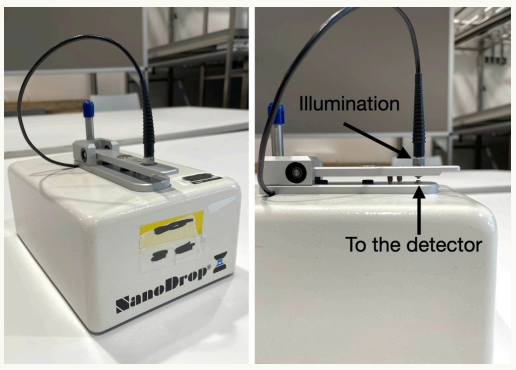
J. VAN LEEUWEN, V. DE BOER, V. SAGGIOMO

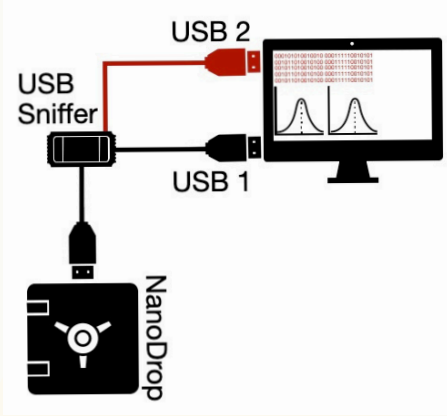
BRINGING OLD LAB INSTRUMENTS BACK TO LIFE

STOP WASTING MONEY AND PRODUCING E-WASTE

01 THE PROBLEM

You own the machine, not the software. The software belongs to the company that made the instrument. And companies don't support their software forever.
After a few years, an instrument gets declared "legacy": no more updates, no more support, no more fixes. Often the software only runs on old computers, which labs can no longer use safely because they're outdated and vulnerable.
So **perfectly good instruments get thrown away, not because they're broken, but because nobody can run them anymore**.
This is wasteful and expensive. It creates tons of e-waste, and it hits teaching labs especially hard, where universities buy many instruments for teaching and then have to discard them while they still work perfectly.



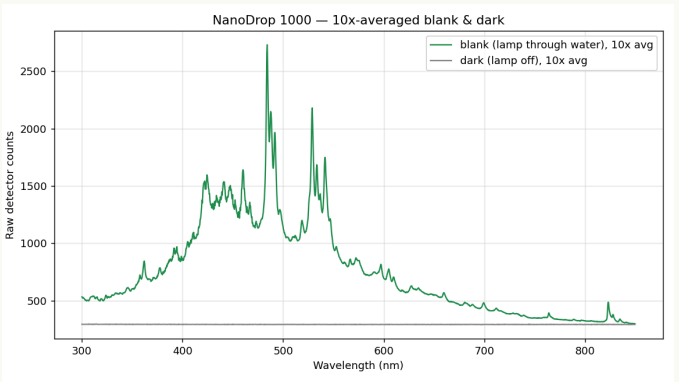


02 WHAT WE DID

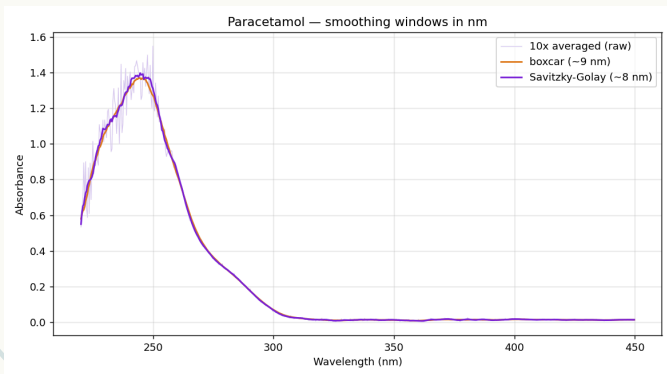
We took an old, unused, ThermoFisher NanoDrop 1000, a small device that measures DNA, RNA, and proteins and brought it back to life.
We reverse-engineered how the instrument "talks" to the computer and we decoded the language it uses.
We wrote our own open-source software in Python to control the instrument.
We connected it to PyLabRobot, an open platform that lets labs automate and control many instruments together from one place.

03 THE OUTCOME

The revived NanoDrop now runs on any modern computer, Windows, Mac, or Linux.
We verified it works correctly. The light spectrum it measures matches the official reference, and it successfully detected paracetamol dissolved in water.
The instrument that was headed for the scrap heap is now a fully working, open piece of laboratory equipment. Everything we made is open-source and free for anyone to use.



04 WHY THIS MATTERS



"Legacy" instruments are usually not broken at all, and, by reviving the software, we can:

- **Save money and reduce e-waste** by keeping working instruments in use instead of buying new ones.
- **Give labs full ownership of their instruments** and their data, free from subscriptions and remote lock-outs.
- **Provide a blueprint for bringing other old instruments back to life**, so this isn't a one-off fix but a repeatable approach.
- **Support open and automated science** by letting instruments talk to each other through open platforms like PyLabRobot.

05 ACKNOWEDGEMENT AND REPOSITORY

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Project funded by WUR Open Science & Education programme (grant no. OSE202534)

OSCW Open Science Community Wageningen

WAGENINGEN UNIVERSITY & RESEARCH

github.com/vcjdeboer/nanodrop-pylabrobot-demo

One instrument at a time, we can turn yesterday's obsolete hardware into tomorrow's open, automated laboratory,and stop throwing working science in the bin.